

IP-SAKTI SAHAYAK

Family 6 — Menispermaceae / Giloy

Objective: Create a TKDL-only family cluster around Giloy so IP-SAKTI can retrieve related plants and rank them as potential alternatives for further research when relevant traditional-knowledge/prior-art or patent risk is detected.

Strict TKDL rule: The official TKDL plant-name index directly lists the three *Tinospora* entries used below. To satisfy the requested 4–5 range without inventing unsupported species, two additional Menispermaceae candidates are included only where the same official TKDL index supports them. Similarity scores are prototype ranking values, not proof of equivalence or legal safety.

1. Anchor / main plant

Giloy — *Tinospora cordifolia* Miers

TKDL Ayurvedic name: Guṛjīrac. Unani: Gilo. Regional names include Seendhil, Amirdhavalli, Somavalli, Kundali, Sagadundam and others.

2. Related TKDL candidates — 5

TKDL plant	Relationship	Similarity*	Evidence level / model use
<i>Tinospora crispa</i>	Same genus; Menispermaceae	VERY HIGH — 0.93	TKDL index; compare specific formulation/use before substitution
<i>Tinospora malabarica</i>	Same genus; Menispermaceae	VERY HIGH — 0.93	TKDL index; compare specific formulation/use before substitution
<i>Cissampelos pareira</i>	Same family; different genus	HIGH — 0.78	TKDL public examination/formulation evidence available
<i>Cocculus hirsutus</i>	Same family; different genus	MEDIUM-HIGH — 0.72	TKDL index; formulation match should be retrieved
<i>Cyclea peltata</i>	Same family; different genus	MEDIUM-HIGH — 0.72	TKDL public examination/formulation evidence available

*Prototype similarity score only. Same genus is a strong screening signal; formulation/use evidence controls the final recommendation.

3. Official TKDL plant-index verification

Official TKDL plant-name index: <https://www.tkdil.res.in/tkdil/langdefault/common/Utility/Keyword/Plant-Name.asp>

The official index lists *Tinospora cordifolia*, *Tinospora crispa* and *Tinospora malabarica* under the *Tinospora* entries. The same TKDL index also contains *Cissampelos pareira*, *Cocculus hirsutus* and *Cyclea peltata*. ■cite■turn0search0■turn0search2■

4. TKDL evidence — Giloy / *Tinospora cordifolia*

4650/MUM/2015: The official TKDL examination report identifies TKDL documents containing *Tinospora cordifolia* in herbal compositions. D1 includes *Tinospora cordifolia* with *Commiphora wightii*, *Baliospermum montanum*, *Embelia ribes* and other ingredients; D2 includes *Tinospora cordifolia* in another multi-ingredient formulation. The report uses these TKDL documents as prior-art material in patent examination.

Official TKDL examination PDF: https://www.tkdil.res.in/tkdil/Outcomes/4650MUM2015_Examiners_report.pdf

1577/DEL/2006: TKDL examination material identifies *Tinospora cordifolia* stem in Ardhavabhedakahara Yoga for hemicrania/migraine and in other traditional formulations. The examiner concluded the relevant claims lacked novelty and inventive step in view of cited TKDL documents.

Official TKDL record: https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/1577_DEL_2006.asp

201621039902: An official Patent Office examination document contains *Tinospora cordifolia* stem in a multi-ingredient therapeutic composition and identifies it among the prior-art ingredients from TKDL-related sources.

Official TKDL examination PDF: https://www.tkdil.res.in/tkdil/Outcomes/201621039902_Examinationreport.pdf

5. Related-species / formulation evidence

***Cissampelos pareira*:** The official TKDL examination material for 2038-MUM-2010 identifies a formulation containing *Cyclea peltata* (*Cissampelos pareira*), *Zingiber officinale*, *Cedrus deodara*, *Cyperus rotundus*, *Tinospora cordifolia*, *Holarrhena antidysenterica*, *Swertia chirayita*, *Picrorhiza kurroa* and *Hemidesmus indicus*. This gives a concrete formulation-level relationship between a same-family candidate and Giloy.

Official TKDL outcome PDF: https://www.tkdil.res.in/tkdil/Outcomes/2038-MUM-2010_FER.pdf

***Cyclea peltata*:** The same official TKDL record provides formulation-level evidence for *Cyclea peltata* (*Cissampelos pareira*) alongside *Tinospora cordifolia*, making it a useful candidate for formulation-similarity retrieval.

***Cocculus hirsutus*:** Retained as a same-family TKDL candidate from the official plant-name index. Do not promote it to a high-confidence substitute until the corresponding TKDL formulation/use evidence is retrieved.

***Tinospora crispa* / *Tinospora malabarica*:** Both are same-genus TKDL entries and therefore receive the highest initial botanical similarity. The model must still retrieve and compare their specific TKDL uses/formulations before recommending them.

6. Alternative-ranking workflow

Example: User has a Giloy-based formulation and IP-SAKTI detects relevant prior-art exposure. The engine identifies *Tinospora cordifolia* → Menispermaceae → retrieves *Tinospora crispa*, *Tinospora malabarica*, *Cissampelos pareira*, *Cocculus hirsutus* and *Cyclea peltata* → compares plant part + indication + formulation role + TKDL evidence → ranks candidates.

Recommended output: “Potential alternative candidate: *Tinospora crispa* — same genus as Giloy, therefore high botanical similarity. Verify traditional-use/formulation overlap and conduct a separate patent/legal search before substitution.”

7. Model-ready schema

family | anchor_plant | candidate_plant | botanical_name | TKDL_name/synonyms | plant_part | traditional_use | formulation | formulation_role | source_text | TKDL_case_id | official_TKDL_url | similarity_score | similarity_reasons | patent_IP_flag | evidence_level | verification_status

8. AI / legal guardrails

- Same family/genus does not establish interchangeability.
- Same family/genus does not establish freedom from patent infringement.
- Never output “no legal barriers” without claim-level patent/legal analysis.
- Separate traditional-knowledge evidence from novelty, inventive step and Section 3(p).
- Show the exact TKDL source behind every recommendation.
- Use “Potential alternative candidate for further research” rather than “safe substitute.”

9. TKDL database limitation

TKDL describes its public representative database as containing about 1,250 formulations from classical Ayurveda, Unani, Siddha and Sowa Rigpa texts. It is searchable by keywords, diseases and IPC codes. This means the public dataset can be used as a verified seed corpus, but it should not be treated as unrestricted access to the complete TKDL corpus. ■cite■turn0search6■